

/// Logging/Recording Data Dictionary

I-ETMS® Protect

Document Type: Data Dictionary
Document Number: WEG-DDG-11079
Revision: 1.4
Date: 08/07/2026
Control Type: CRB Controlled
Distribution Type: Limited External

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/// Revision History

Date	Revision	Description	Author
	0.1	This data dictionary is based off revision 2.1 of WCR-SYR-1057.	
12/1/2022	1.0	DCR 174896 - initial revision for I-ETMS Protect DCR 160439 - updates for supporting other languages (UTF-8 characters) DCR 169391 - updated LID and added CSI record DCR 177662 - updated GPS record	Wabtec
4/24/2023	1.1	DCR 196517 - update LID record for Control System Subtype	Wabtec
4/23/2024	1.2	DCR 178626 - HOTCC Interface DCR 208490 - Logging of EOD ITF/BCP Overlay removal DCR 175499 - Rear End Position Adjustment from PTC-EOT impacts on Automatic Rollup of Authorities and Train Integrity DCR 221844 - HOTCC Data Status Record example text fix DCR 234614 - (231660/230698) - updated SYS record	Wabtec
6/30/2025	1.3	DCR 263098 (262675/174275) - add in L2Is for logging of Energy Management messages DCR 263113 (202210) - add fields in L2I for Function Supported and Operating State	Wabtec
8/7/2026	1.4	DCR 293557 (159871) - Restore reason for state change to include Penalty Switch Cutout DCR 380972 (331056) - Add in L2I for Train Handling Data Logging feature DCR 292357 (180412) - Update the description of the FLT record for L2I431 DCR 293768(128773(102926/17778)) - Update TRN Record DCR 293369 (160997) - Added a new Reason for State Change (PTC Suspension) DCR 383874 (383853) - Updated for the Train Handling Data Logging feature DCR 383860 (383466) - Updated for the Train Handling Data Logging feature DCR 294027 (159860) - Updated for the SR KSS EMU DCR 268588 (170660) - Updated Key Display Record example DCR 391553 - EM: Update EOD, ECI, ESP Log versions to 2 DCR 247475 (165852) - Update functionality regarding switches in PTC suspension limits DCR 387633 - Update TGT record header DCR 386526 - MDR record updates DCR 401325 (103026) - Addition of AUD and MNU DCR 293877 - Update LOC, TRK, FDS, TGT, TDV, and EXC for SCR-5529 / GAP-051 DCR 382773 - Update TGT for SCR-5531 / GAP-002 DCR 405272 - Update AUD for adjusted enumeration	Wabtec

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/// 1 Introduction

1.1 Purpose

The purpose of this data dictionary is to define the information and format the I-ETMS application logs and records on-board the locomotive.

1.2 Scope

The scope of this document is limited to the information the I-ETMS application logs and records on-board.

1.3 References

The following documents are referenced by this data dictionary:

- I-ETMS® Protect On-Board Segment Requirement Specification, WEG-SYR-11066. Wabtec.
(See *I-ETMS Protect Compatibility and Compliance Matrix* for applicable version)
- I-ETMS® Protect Component, Fault, and Event Data Dictionary, WEG-DDG-11078. Wabtec.
(See *I-ETMS Protect Compatibility and Compliance Matrix* for applicable version)

1.4 Acronyms

Acronym	Description
AG	Application Gateway
CHM	Crash Hardened Memory
EM	Energy Management
EMP	Edge Message Protocol
FBAR	Form Based Authority Required
GPS	Global Positioning System
HDOP	Horizontal Dilution of Precision
MDM	Mobile Device Manager
MP	Milepost
PPR	Pulses Per Revolution
PTC	Positive Train Control
SCAC	Standard Carrier Alpha Code
THE	Train Handling Exception
UTC	Coordinated Universal Time
WIU	Wayside Interface Unit

1.5 Document Organization

This document is organized into three sections:

- **Section 1 – Introduction**, defines the purpose and scope, identifies referenced documents, provides definitions for key elements and acronyms, and provides an organizational overview of the standards guide.
- **Section 2 – Log File Name Format**, defines the log file name format.
- **Section 3 – Data Records**, provides the information and format of the Record Header and Data Records defined for logging and recording.

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/// 2 Log File Name Format

This section defines the log file name format used by the on-board segment. The log file name contains a UTC timestamp of the last event logged in the file.

The following structure defines the on-board segment log file name format:

chr.RRRR.NNNNNN.YYYYMMDDHHMMSS.log.gz

chr = log file name, lower case

RRRR = SCAC, 1 - 4 Alpha, upper case

NNNNN = Locomotive Number, 1 – 6 Alphanumeric, if alpha then upper case

YYYY = Year, 4 Numeric

MM = Month, 01 - 12

DD = Day, 01 - 31

HH = Hour, 00 - 23

MM = Minute, 00 - 59

SS = Second, 00 - 59

.log = log file type, lower case

.gz = file compressed, lower case

Examples when Loco ID is known:

chr.RR.1234.20111213141516.log.gz

chr.RRRR.567890.20120102030405.log.gz

Example when Loco ID is unknown:

chr.LOCO.XXXX.20111212301530.log.gz

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/// 3 Data Records

This section defines the record header and data records for each type of record to be logged and recorded.

3.1 Record Header

L2I408

The record header is attached to the beginning of each data record that is logged or recorded.

L2R408.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I408.1.5 UTC Date	10	Fixed	yyyy/mm/dd
L2I408.1.3 UTC Time	12	Fixed	hh:mm:ss.sss , applied when event is logged.
L2I408.1.4 Record ID	3	Variable	<see Record ID and Record Name Table>
L2I408.1.2 Record Name	3	Variable	<see Record ID and Record Name Table>
L2I408.1.1 Version	2	Variable	<defined in heading of each Data Record Type>

3.1.1 Record ID and Record Name Table

L2I408.2

The following table provides the Record ID and Record Name for each record type.

L2I408.3

The Record ID is a numerical representation of the Record Name.

L2I408.4

When a record is logged or recorded, the associated Record ID and Record Name is populated in the record header for that record type.

L2I408.5 ID	Name	Data Record Type
L2I408.5.14 1	SYS	System Record
L2I408.5.10 2	CFG	Configuration Record
L2I408.5.16 3	TRN	Train Information Record
L2I408.5.8 4	DEP	Departure Test Record
L2I408.5.6 5	GPS	GPS Record
L2I408.5.4 6	LOC	Location Record
L2I408.5.11 7	TRK	Track Type Record
L2I408.5.15 8	SYN	Synchronization Record
L2I408.5.19 9	FDS	Field Device Status Record
L2I408.5.9 10	CSS	Cab Signal Status Record
L2I408.5.17 11	MDR	Mandatory Directive Record
L2I408.5.20 12	BNR	Banner Record
L2I408.5.21 13	PRM	Prompt Record
L2I408.5.22 14	KDP	Key Displayed Record
L2I408.5.1 15	KPR	Key Pressed Record
L2I408.5.5 16	TGT	Target Record
L2I408.5.3 17	BHA	Brake and Horn Application Record
L2I408.5.2 18	THE	Train Handling Exception Record
L2I408.5.12 19	FLT	Fault Record
L2I408.5.18 20	LID	Locomotive ID Record
L2I408.5.7 21	EMP	EMP Message Record
L2I408.5.13 22	EMF	EMP Message Failure Record
L2I408.5.23 23	TDV	Track Data Version Record

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L2I408.5.24	24	Reserved	Reserved
L2I408.5.25	25	Reserved	Reserved
L2I408.5.26	26	Reserved	Reserved
L2I408.5.27	27	EXC	Exclusion Zone
L2I408.5.28	28	ECI	Energy Management Control Indicator
L2I408.5.29	29	EOD	Energy Management Overlay Display
L2I408.5.30	30	ESP	Energy Management Speed Prediction
L2I408.5.31	31	EMI	Energy Management Information
L2I408.5.32	32	CSI	Serial Cab Signal System Interface
L2I408.5.33	33	EGP	EOT GPS
L2I408.5.34	34	EST	EOT Status
L2I408.5.35	35	HDS	HOTCC Data Status
L2I408.5.36	36	Reserved	Reserved
L2I408.5.37	37	PBK	Phase Break
L2I408.5.38	38	Reserved	Reserved
L2I408.5.39	39	AUD	Audible Alert Record
L2I408.5.40	40	MNU	Menu Selection
L2I408.5.41	41	THD	Train Handling Data

3.2 System (SYS) – Version 6

L2I409

Log and record the data record when a data field changes status.

Record Content:

L2I409.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I409.1.2 On-board State	2	Fixed	SB: Start Boot ST: Self Test CO: Cut Out IN: Initializing DS: Disengaged SW: Restricted AC: Active FL: Failed
L2I409.1.7 Reason for State Change	2	Variable	0: No Change 1: Pass Self Test 2: RESERVED (Not Used) 3: Crew Selected Init 4: Crew Selected Cutin 5: Crew Selected Cutout 6: Crew Selected Logoff 7: Office Commanded Cutout 8: Air Brake Cutout 9: Penalty Switch Cutout 10: Successful Init and Departure Test 11: Unsuccessful Departure Test 12: Crew Exit Initialization 13: RESERVED (Not Used) 14: RESERVED (Not Used) 15: Conditions for Active State Satisfied 16: Direction of Travel Unknown 17: Location Unknown 18: Exit PTC Track 19: RESERVED (Not Used) 20: Subdivision Non-Synchronized 21: Crew Enter Restricted State 22: Exit Restricted State

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L2I409.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
			23: Failure Detected 24: Failure Cleared 25: Crew exit initialization – request timed out 26: Crew exit initialization – system configuration incompatible 27: Crew exit initialization – software download 28: Crew exit initialization – verification failure 29: Crew exit initialization – no train ID 30: Crew exit initialization – system configuration invalid 31: Software installation timeout 32: Request for configuration timed out 33: Invalid configuration 34: Automatically Enter Restricted State 35: Reverse train orientation on unknown switch 36: Enter PTC suspension 37: Exclusion zone needed
L2I409.1.4 Penalty Switch	1	Fixed	0: Cut Out 1: Cut In
L2I409.1.8 Emergency Switch	1	Fixed	0: Cut Out 1: Cut In
L2I409.1.5 Horn Switch	1	Fixed	0: Cut Out 1: Cut In
L2I409.1.3 Air Brake	1	Fixed	0: Cut Out 1: Cut In
L2I409.1.11 Cab Signal	1	Fixed	0: Cut Out 1: Cut In
L2I409.1.6 Park Mode	1	Fixed	0: Disabled 1: Enabled
L2I409.1.13 AG I/F	1	Fixed	0: Disconnected 1: Connected
L2I409.1.10 CHM I/F	1	Fixed	0: Disconnected 1: Connected
L2I409.1.9 EM I/F	1	Fixed	0: Disconnected 1: Connected
L2I409.1.1 MDM I/F	1	Fixed	0: Disconnected 1: Connected

Example of System Record:

2011/06/05|12:00:00.000|1|SYS|6|CO|3|1|1|1|1|0|0|1|1|1|1

Note: Shaded: record header, Non-shaded: data record

3.3 Configuration (CFG) – Version 2

L2I410

Log and record the data record at system reboot and when a new file is installed.

Record Content:

L2I410.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I410.1.8 Repository	50	Variable	Repository Name

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L2I410.1.9 Fileset Name	50	Variable	Fileset Name
L2I410.1.10 Fileset Version	20	Variable	Fileset Version Name
L2I410.1.11 Checksum	8	Fixed	First 8 Hexadecimal Characters of the MD5 Hash of Fileset
L2I410.1.12 Fileset Status	1	Fixed	0: Not used 1: Preferred 2: Acceptable

Example of Configuration Record:

2011/06/05|12:00:00.000|2|CFG|2|Software|WPN27185 I-ETMS|6.1.2.1|cae283e9|1

Note: Shaded: record header, Non-shaded: data record

3.4 Train Info (TRN) – Version 5

L2I411

Log and record the data record when a data field changes status and each time a new log file is created.

L2I412

This record should be the first record logged in a new file.

Record Content:

L2I411.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I411.1.18 Loco ID	10	Variable	Locomotive Initial and Number.
L2I411.1.24 Number of Train IDs	2	Fixed	0-5
L2I411.1.25 For each Train ID			Repeated for each Operating Railroad selected at initialization Train IDs
L2I411.1.19 Train ID	30	Variable	Train ID selected at Initialization
L2I411.1.26 Operating SCAC	4	Variable	Operating Railroad associated with each instance of Train ID
L2I411.1.27 Clearance No	8	Variable	Clearance number entered at Initialization
L2I411.1.20 Employee ID	8	Variable	Employee ID entered at Initialization, Note: if Employee ID is not available, field will be empty
L2I411.1.21 Employing SCAC	4	Variable	Employing Railroad. Note: if Employing SCAC is not available, field will be empty
L2I411.1.14 Employee Logon Status	1	Fixed	0: No Employee Logged On 1: Unsuccessful 2: Successful
L2I411.1.22 Clearance No.	8	Variable	Clearance number entered at Initialization, Note: if Clearance Number is not available, field will be empty
L2I411.1.13 Train Type	2	Fixed	CM: Commuter FR: Freight HS: High Speed Passenger IM: Intermodal PS: Passenger TT: Tilt Train
L2I411.1.15 Loco Orientation	1	Fixed	0: Unknown F: Front B: Back
L2I411.1.7 Loco Count	2	Variable	1 - 24
L2I411.1.12 Trailing Tonnage	5	Variable	0 - 45000 tons.
L2I411.1.11 Op Brake Count	3	Variable	0 - 999
L2I411.1.6 Axle Count	4	Variable	0 - 3996

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L2I411.1.17 Total Length	5	Variable	0 - 25000 feet.
L2I411.1.23 Loaded Car Count	3	Variable	0 - 999
L2I411.1.10 Empty Car Count	3	Variable	0 - 999
L2I411.1.4 Brake Force	8	Variable	0 - 30000000 pounds
L2I411.1.5 Equip Speed	3	Variable	0 - 125 mph
L2I411.1.28 Train Restriction Type Count	2	Variable	0 - 32 Number of train restriction types in this record, repeat following fields for each train restriction type
L2I411.1.29 Train Restriction Type Code	2	Variable	1 - 32 Numeric code assigned to train restriction type in summary consist and track data
L2I411.1.30 Train Restriction Type Description	30	Variable	Textual description of train restriction type

Example of Train Info Record:

2011/06/05|12:00:00.000|3|TRN|5|RR2002|1|RR1AX01|RR|5678|12345678|RR|2|56748|FR|F|2|6250|25|100|1700|25|0|6250000|60|1|1|Train Restriction Type Description

Note: Shaded: record header, Non-shaded: data record

3.5 Departure Test (DEP) – Version 4

L2I413

Log and record the data record at completion of departure test.

Record Content:

L2I413.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I413.1.2 Latitude	10	Variable	dd:mm.mmmm
L2I413.1.14 N/S Indicator	1	Fixed	I: Invalid N: North S: South
L2I413.1.4 Longitude	11	Variable	ddd:mm.mmmm
L2I413.1.15 E/W Indicator	1	Fixed	I: Invalid E: East W: West
L2I413.1.7 Departure Test Location	45	Variable	Name of location where the departure test was ran, Note: if no location available than field will be empty
L2I413.1.8 Employee ID	8	Variable	Employee ID of employee who performed departure test
L2I413.1.9 Employing SCAC	4	Variable	Employing SCAC of employee who performed departure test
L2I413.1.10 Penalty Brake Pass/Fail Status	1	Fixed	F: Failed P: Passed
L2I413.1.11 Alerter Pass/Fail Status	1	Fixed	F:Failed P:Passed
L2I413.1.13 Test Time Stamp	14	Fixed	Test completion date and time YYYYMMDDHHMMSS

Example of Departure Test Record:

2011/06/05|12:00:00.000|4|DEP|4|33:24.7020|N|80:50.5514|W|ABC Yard|12345678|RR|P|P|20120320123045

Note: Shaded: record header, Non-shaded: data record

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3.6 GPS (GPS) – Version 4

L2I414

Log and record the data record every 30 seconds when stopped or moving without a track location and every TBC112 (5 seconds) when moving with a track location.

L2I415

Also log when Position Fix changes state.

Record Content:

L2I414.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I414.1.8 GPS Source	1	Fixed	1: GPS 1 2: GPS 2
L2I414.1.5 Latitude	10	Variable	dd:mm.mmmm
L2I414.1.6 N/S Indicator	1	Fixed	I: Invalid N: North S: South
L2I414.1.3 Longitude	11	Variable	ddd:mm.mmmm
L2I414.1.4 E/W Indicator	1	Fixed	I: Invalid E: East W: West
L2I414.1.2 Position Fix	1	Fixed	0: Invalid 1: Valid SPS/Standalone 2: Valid DGPS/SBAS 3: Valid PPS 4: RTK
L2I414.1.1 Number of Satellites	2	Fixed	00 - 999, total number of satellites across all constellations
L2I414.1.7 HDOP	4	Variable	nn.n

Example of GPS Record:

2011/06/05|12:00:00.000|5|GPS|4|1|33:24.7020|N|80:50.5514|W|2|007|1.4

Note: Shaded: record header, Non-shaded: data record

3.7 Location (LOC) – Version 6

L2I416

Log and record the data record every 30 seconds when stopped and every 1 second when moving.

Record Content:

L2I416.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I416.1.1 Current Speed	6	Variable	nnn.nn mph
L2I416.1.5 Max Speed Limit	3	Variable	Mph, Note: field empty if leading edge location is unknown
L2I416.1.7 Leading Edge Location Operating Railroad SCAC	4	Variable	Railroad SCAC of the railroad that operates the leading edge subdivision/district, Note: field empty if leading edge location is unknown
L2I416.1.8 Leading Edge Location Sub/Dist ID	5	Variable	Current subdivision/district of the leading edge location, Note: field empty if leading edge location is unknown
L2I416.1.9 Leading Edge Location Track Name	20	Variable	Head end track name, Note: field empty if leading edge location is unknown

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L2I416.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I416.1.10 Leading Edge Location MP	19	Variable	pppppxxxx.xxxxxssss, Note: field empty if leading edge location is unknown
L2I416.1.11 Leading Edge Location Block ID	10	Variable	Track Subdiv Block ID, Note: field empty if leading edge location is unknown
L2I416.1.12 Leading Edge Location Offset	6	Variable	Foot offset into Track Subdiv Block ID, Note: field empty if leading edge location is unknown
L2I416.1.21 Leading Edge Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I416.1.13 Trailing Edge Location Operating Railroad SCAC	4	Variable	Railroad SCAC of the railroad that operates the trailing edge subdivision/district, Note: field empty if trailing edge location is unknown
L2I416.1.14 Trailing Edge Location Sub/Dist ID	5	Variable	Current subdivision/district of the trailing edge location, Note: field empty if trailing edge location is unknown
L2I416.1.15 Trailing Edge Location Track Name	20	Variable	Trailing Edge track name, Note: field empty if trailing edge location is unknown
L2I416.1.16 Trailing Edge Location MP	19	Variable	pppppxxxx.xxxxxssss, Note: field empty if trailing edge location is unknown
L2I416.1.17 Trailing Edge Location Block ID	10	Variable	Track Subdiv Block ID, Note: field empty if trailing edge location is unknown
L2I416.1.18 Trailing Edge Location Offset	6	Variable	Foot offset into Track Subdiv Block ID, Note: field empty if trailing edge location is unknown
L2I416.1.22 Trailing Edge Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I416.1.4 Direction of Travel	1	Fixed	0: Unknown 1: Increasing MP 2: Decreasing MP
L2I416.1.2 Position Uncertainty	4	Variable	0 - 1000 meters, position uncertainty for leading edge of train
L2I416.1.19 Calibrated Wheel Diameter	5	Variable	nn.nn inches
L2I416.1.20 Trailing Edge Position Uncertainty	4	Variable	0 - 1000 meters, position uncertainty for trailing edge of train

Example of Location Record:

2011/06/05|12:00:00.000|6|LOC|6|35.03|40|RR|321|Main|20.5000|1103|15063|US|RR|321|Main|20.2000|1103|13763|US|1|5|40.25|5

Note: Shaded: record header, Non-shaded: data record

3.8 Track Type (TRK) – Version 3

L2I417

Log and record the data record when a data field changes status. Note: The PTC Type and Method of Operation applies to the location currently occupied by the leading edge of the train inside the Country of Operation.

Record Content:

L2I417.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
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L2I417.1.1 PTC Type	1	Fixed	1: PTC Territory 2: Non PTC Territory 3: PTC Entry
L2I417.1.8 Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I417.1.4 Method of Operation Signal Direction	1	Fixed	1: Increasing 2: Decreasing 3: Bidirectional 4: None
L2I417.1.5 Method of Operation FBAR	1	Fixed	1: Increasing MP 2: Decreasing MP 3: Bidirectional 4: None
L2I417.1.6 Method of Operation Cab Signal	1	Fixed	1: None 2: Route Based 3: Speed Enforcement
L2I417.1.7 Method of Operation Yard Limits	1	Fixed	1: True 2: False

Example of Track Type Record:

2011/06/05|12:00:00.000|7|TRK|3|1|US|4|3|1|2

Note: Shaded: record header, Non-shaded: data record

3.9 Synchronization (SYN) – Version 2

L2I418

Log and record the data record when a data field changes status.

Record Content:

L2I418.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I418.1.6 SCAC	4	Variable	SCAC of the Subdivision/District included in Subdivision/District
L2I418.1.5 PTC Sub/Dist ID	5	Variable	ID of Subdivision/District included in Subdivision/District list
L2I418.1.2 Synchronized	1	Fixed	0: Non Synchronized 1: Synchronized
L2I418.1.1 Poll Rate Registration	1	Fixed	F: Fast M: Medium S: Slow
L2I418.1.3 Dataset CRC-32	8	Fixed	Aggregate CRC of Dataset .

Example of Synchronization Record:

2011/06/05|12:00:00.000|8|SYN|2|RR|321|1|F|63568fe9

Note: Shaded: record header, Non-shaded: data record

3.10 Field Device Status (FDS) – Version 6

L2I419

Log and record the data record when a data field changes status.

Record Content:

L2I419.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
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L2I419.1.1 Device Type	2	Fixed	WI: Monitored Signal Indication WS: Monitored Switch Position WD: Monitored Hazard Detector UI: Unmonitored Signal Indication US: Unmonitored Switch Position
L2I419.1.4 WIU UTC Creation Time	14	Fixed	yyyymmddhhmmss
L2I419.1.7 WIU ID	12	Fixed	12 decimal digit string WIU ID of monitored device as defined in the ITC Wayside Locomotive ICD, blank if record is for unmonitored device
L2I419.1.9 Device SCAC	4	Variable	SCAC of the device
L2I419.1.10 Device Sub/Dist ID	5	Variable	Subdivision/District ID of the device
L2I419.1.11 Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I419.1.8 Device ID	10	Variable	Device ID for both monitored and unmonitored devices
L2I419.1.3 Device Indication	3	Fixed	Unk: Unknown Nrm: Normal Switch Position Rev: Reverse Switch Position DRK: Dark SG1: Signal Group 1 Signal Indication SG2: Signal Group 2 Signal Indication SG3: Signal Group 3 Signal Indication SG4: Signal Group 4 Signal Indication SG5: Signal Group 5 Signal Indication NCD: No Condition Detected CDD: Condition Detected
L2I419.1.2 Reason for Change	2	Variable	0: Invalid 1: Received Monitored Status Message 2: Reserved 3: Status Timed Out 4: Crew Indicates Switch Position 5: Signal Status set by Cab Signal 6: Unmonitored Switch governed by Signal 7: Unmonitored Switch monitored by Track Circuit 8: Unmonitored Signal 9: PTC suspension 10: On-board Determines Switch Position

Example of Field Device Status Record:

2011/06/05|12:00:00.000|9|FDS|6|WI|20110506115959|780212345678|RR|321|US|6061|SG1|1

Note: Shaded: record header, Non-shaded: data record

3.11 Cab Signal Status (CSS) – Version 4

L2I420

Log and record the data record when a data field changes status.

Record Content:

L2I420.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I420.1.2 Cab Signal Indication	7	Variable	INV: Invalid Cab Signal KDO: Known Drop Out Location 0: Cab Code Rate 0 75: Cab Code Rate 75

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			75.75: Cab Code Rate 75/75 120: Cab Code Rate 120 270: Cab Code Rate 270 120.120: Cab Code Rate 120/120 270.270: Cab Code Rate 270/270 180: Cab Code Rate 180 180.180: Cab Code Rate 180/180
--	--	--	--

Example of Cab Signal Status Record:

2011/06/05|12:00:00.000|10|CSS|4|75

Note: Shaded: record header, Non-shaded: data record

3.12 Mandatory Directive (MDR) – Version 3

L2I421

Log and record the data record when a mandatory directive status changes.

Record Content:

L2I421.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I421.1.1 Operating Railroad SCAC	4	Variable	Railroad SCAC of railroad that issued mandatory directive
L2I421.1.4 Type	2	Fixed	TW: Track Authority / Track Warrant TT: Track Permit / Track & Time ET: Enter Main Track PS: Pass Signal at Stop SR: Speed Restriction WZ: Work Zone GC: Grade Crossing TS: Track Out of Service AC: Advisory/ Cautionary CA: Critical Alert SU: PTC Suspension PE: CROR PSS Equipment PN: CROR PSS No Equipment
L2I421.1.2 Reference No.	10	Variable	Mandatory directive's reference number
L2I421.1.3 Status	1	Fixed	P: Pending A: Active R: Releasing V: Void D: Discarded

Example of Mandatory Directive Record:

2011/06/05|12:00:00.000|11|MDR|3|RR|TW|56789|A

Note: Shaded: record header, Non-shaded: data record

3.13 Banner (BNR) – Version 1

L2I422

Log and record the data record when a banner is displayed and when the banner is removed.

Record Content:

L2I422.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I422.1.2 Banner Status	1	Fixed	D: Displayed R: Removed

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L2I422.1.1 Banner Text	30	Variable	Banner text (UTF-8)
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Example of Banner Record:

2011/06/05|12:00:00.000|12|BNR|1|D|Braking in Progress

Note: Shaded: record header, Non-shaded: data record

3.14 Prompt (PRM) – Version 2

L2I423

Log and record the data record when a prompt is displayed, overlaid by another screen or banner, and when the prompt is removed by key press, message, event or time out.

Record Content:

L2I423.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I423.1.1 Prompt Status	1	Fixed	D: Displayed; log when banner first appears or reappears O: Overlaid; log when overlaid by another screen or banner R: Removed; log when removed by key press, message, event, or time out
L2I423.1.3 Source Code	3	Fixed	PTC: Positive Train Control EMS: Energy Management System
L2I423.1.2 Prompt Text	180	Variable	Prompt text (UTF-8)

Example of Prompt Record:

2011/06/05|12:00:00.000|13|PRM|2|D|PTC|Speed Reduction to 30 MPH

Note: Shaded: record header, Non-shaded: data record

3.15 Key Display (KDP) – Version 3

L2I424

Log and record the data record when the key menu changes.

Record Content:

L2I424.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I424.1.9 Source Code	3	Fixed	PTC: Positive Train Control EMS: Energy Management System
L2I424.1.2 Key 1 Text	30	Variable	Text displayed at soft key position 1 (UTF-8)
L2I424.1.6 Key 2 Text	30	Variable	Text displayed at soft key position 2 (UTF-8)
L2I424.1.1 Key 3 Text	30	Variable	Text displayed at soft key position 3 (UTF-8)
L2I424.1.4 Key 4 Text	30	Variable	Text displayed at soft key position 4 (UTF-8)
L2I424.1.3 Key 5 Text	30	Variable	Text displayed at soft key position 5 (UTF-8)
L2I424.1.5 Key 6 Text	30	Variable	Text displayed at soft key position 6 (UTF-8)
L2I424.1.7 Key 7 Text	30	Variable	Text displayed at soft key position 7 (UTF-8)
L2I424.1.8 Key 8 Text	30	Variable	Text displayed at soft key position 8 (UTF-8)
L2I424.1.10 Text	4000	Variable	Display Format Text where each line is terminated by the LF (0x0A) character and attributes and position of the text can be specified using the ANSI standard terminal control sequences.

Note: The following graphic keys are represented by the associated text.

Graphic	CHR Record Content
---------	--------------------

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Key	
	[DOWN]
	[UP]
	[LEFT]
	[RIGHT]
	[SWITCH_RIGHT]
	[SWITCH_LEFT]

Example of Key Display Record:

2011/06/05|12:00:00.000|14|KDP|3|PTC|Mandatory Directives|Consist||Restricted Mode On||||Menu 1

Note: Shaded: record header, Non-shaded: data record.

3.16 Key Pressed (KPR) – Version 2

L2I426

Log and record the data record when a key is pressed,

L2I427

for functions that require 2 simultaneous keys pressed at the same time, enter the function name (i.e., diagnostics and track validation)

Record Content:

L2I426.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I426.1.2 Source Code	3	Fixed	PTC: Positive Train Control EMS: Energy Management System
L2I426.1.1 Key Text	30	Variable	Text of pressed key

Example of Key Pressed Record:

2011/06/05|12:00:00.000|15|KPR|2|PTC|Consist

Note: Shaded: record header, Non-shaded: data record

3.17 Target (TGT) – Version 6

L2I428

Log and record the data record for enforceable targets when system is in Active state - when content of the record has changed and every 60 seconds when the content of the record has not changed.

Record Content:

L2I428.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I428.1.4 Warning Distance	6	Variable	Feet Displayed on CDU
L2I428.1.2 Braking Distance	6	Variable	Feet Displayed on CDU

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L2I428.1.5 Next Target Distance	6	Variable	Feet Displayed on CDU
L2I428.1.1 Enforceable Target Count	3	Variable	0-100 Number of targets in this record, repeat following fields for each target. There is no order to the targets listed in this record.
L2I428.1.8 Start Location Operating Railroad SCAC	4	Variable	SCAC of Operating Railroad of the subdiv of the target's starting location
L2I428.1.9 Start Location Sub/Dist ID	5	Variable	Track Subdivision / District ID of the target's starting location
L2I428.1.10 Start Location Block ID	10	Variable	Track Segment Block ID of the target's starting location
L2I428.1.11 Start Location Offset	6	Variable	Foot offset into Block of the target's starting location
L2I428.1.20 Start Location Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I428.1.12 End Location Operating Railroad SCAC	4	Variable	SCAC of Operating Railroad of the subdiv of the target's ending location
L2I428.1.13 End Location Sub/Dist ID	5	Variable	Track Subdivision / District ID of the target's ending location
L2I428.1.14 End Location Block ID	10	Variable	Track Segment Block ID of the target's ending location
L2I428.1.15 End Location Offset	6	Variable	Foot offset into Block of the target's ending location
L2I428.1.21 End Location Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I428.1.16 Target Type	80	Variable	Type of Target
L2I428.1.17 Target Speed Limit	3	Variable	mph Target speeds of 600 mph denote targets that are non-enforceable and used for track line colorization only.
L2I428.1.18 Speed Applicability	1	Fixed	0 = Entire Train 1 = Head End Only
L2I428.1.19 Restricted Speed	1	Fixed	0 = Not Restricted Speed 1 = Restricted Speed
L2I428.1.22 Reduced Speed	1	Fixed	0 = Not Reduced Speed 1 = Reduced Speed

Example of Target Record:

2011/06/05|12:00:00.000|16|TGT|6|2500|1500|5000|1|RR|321|1103|10050|US|RR|321|1104|65|US|Work
Zone|49|1|0|0

Note: Shaded: record header, Non-shaded: data record.

3.18 Brake and Horn App (BHA) – Version 3

L2I429

Log and record the data record every one second from when the brakes are applied to when the train is stopped or from when the horn sounds to when the horn stops sounding.

Record Content:

L2I429.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I429.1.1 Brake Pipe Pressure	3	Variable	psi
L2I429.1.3 Eq Res Pressure	3	Variable	psi
L2I429.1.4 Penalty Application	1	Fixed	0: No Penalty 1: PTC Induced
L2I429.1.2 Emergency Application	1	Fixed	0: No Emergency 1: PTC Induced

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			2: Engineer Induced 3: Conductor Induced 4: Train Induced
L2I429.1.5 Horn Application	1	Fixed	0: No Horn 1: PTC Induced 2: Engineer Induced
L2I429.1.6 DOT ID	7	Variable	DOT ID of the crossing associated with the horn application. Blank if not applicable.

Example of Brake and Horn App Record:

2011/06/05|12:00:00.000|BHA|3|90|90|0|0|1|1234XYZ

Note: Shaded: record header, Non-shaded: data record

3.19 Train Handling Exception (THE) – Version 2

L2I430

Log and record the data record when a train handling exception is detected and when it clears.

Record Content:

L2I430.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I430.1.1 Train Handling Exception	20	Variable	Textual description of Train Handling Exception
L2I430.1.2 THE Status	1	Fixed	0: Cleared 1: Detected

Example of Train Handling Exception Record:

2011/06/05|12:00:00.000|18|THE|2|Power Braking|1

Note: Shaded: record header, Non-shaded: data record

3.20 Fault (FLT) – Version 2

L2I431

Log and record the data record when a fault or event status changes.

Record Content:

L2I431.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I431.1.2 Fault Status	1	Fixed	0: Cleared 1: Active 2: Intermittent
L2I431.1.5 Fault Source	20	Variable	Source of the named Component that raised or cleared fault, provided in I-ETMS Protect Component, Fault, and Event Data Dictionary, or the source of the event.
L2I431.1.4 Fault Code	4	Fixed	Unique code number for the fault or event, provided in I-ETMS Protect Component, Fault, and Event Data Dictionary.
L2I431.1.1 Reported Fault	64	Variable	Fault or event description, provided in I-ETMS Protect Component, Fault, and Event Data Dictionary

Example of Fault Record:

2011/06/05|12:00:00.000|19|FLT|2|1|LocoID|05fa|Test Fail

Note: Shaded: record header, Non-shaded: data record

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3.21 Loco ID (LID) – Version 8

L2I432

Log and record the data record when the on-board system boots up or when the Loco ID Setting changes status.

Record Content:

L2I432.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I432.1.20 Loco Id Setting	1	Fixed	0: Unlocked 1: Locked
L2I432.1.13 Reporting Mark SCAC	4	Variable	SCAC stenciled on side of locomotive
L2I432.1.12 Engine Number	6	Variable	Number
L2I432.1.17 Owning/Leasing SCAC	4	Variable	Railroad SCAC of railroad that owns or leases locomotive
L2I432.1.15 GPS 1 Ant Front Offset	4	Variable	0 - 1800 inches
L2I432.1.19 GPS 1 Ant Center Offset	2	Variable	0 - 60 inches
L2I432.1.8 GPS 1 Ant Center Reference	1	Fixed	L: Left R: Right
L2I432.1.9 GPS 2 Ant Front Offset	4	Variable	0 - 1800 inches
L2I432.1.6 GPS 2 Ant Center Offset	2	Variable	0 - 60 inches
L2I432.1.14 GPS 2 Ant Center Reference	1	Fixed	L: Left R: Right
L2I432.1.10 Loco Weight	3	Variable	100 - 800 thousand pounds
L2I432.1.7 Loco Length	3	Variable	40 - 150 feet
L2I432.1.5 Loco Horsepower	5	Variable	0 - 10,000 HP
L2I432.1.11 Loco Traction	2	Fixed	AC: AC traction DC: DC traction
L2I432.1.16 Number of Loco Axles	2	Variable	4: 4 axle locomotive 6: 6 axle locomotive 8: 8 axle locomotive 12: 12 axle locomotive 28: 28 axle locomotive
L2I432.1.1 Wheel Diameter	5	Variable	6.00 - 46.00 inches
L2I432.1.2 Wheel Tach PPR	4	Variable	6 - 1200
L2I432.1.18 Cab Signal Code 0	1	Fixed	0: Not Connected 1: Connected
L2I432.1.26 Cab Signal Code 75	1	Fixed	0: Not Connected 1: Connected
L2I432.1.27 Cab Signal Code 120	1	Fixed	0: Not Connected 1: Connected
L2I432.1.28 Cab Signal Code 180	1	Fixed	0: Not Connected 1: Connected
L2I432.1.29 Cab Signal Code 270	1	Fixed	0: Not Connected 1: Connected
L2I432.1.30 Cab Signal Code 75/75	1	Fixed	0: Not Connected 1: Connected
L2I432.1.31 Cab Signal Code 120/120	1	Fixed	0: Not Connected 1: Connected
L2I432.1.32 Cab Signal Code 180/180	1	Fixed	0: Not Connected 1: Connected
L2I432.1.33 Cab Signal Code	1	Fixed	0: Not Connected

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270/270			1: Connected
L2I432.1.22 Control System Type	2	Variable	0: ACP 1: ACP_NRZI 2: LSI 3: Reserved 4: Reserved 5: LSI_EPIC_QES 6: TMC_DIO 7: LIG 8: LSI_ASYNC 9: Reserved 50: HR_EMU 51: NS_HL2_EMU 52: CRM_EMU 53: NS_MC_EMU 54: NS_MCEW_EMU 55: NS_300_EMU 56: SR_FLRT3_DMU 57: SR_KSS_EMU Note: the enumeration is a profile of the various locomotive configurations the TMC can interface with, only one enumeration can be set per locomotive configuration
L2I432.1.3 CPU 4	1	Fixed	0: No 1: Yes
L2I432.1.4 CPU 10	1	Fixed	0: No 1: Yes
L2I432.1.23 Crash Hardened Memory	1	Fixed	0: No 1: Yes
L2I432.1.34 Serial Cab Signal System Interface	1	Fixed	0: No 1: Yes
L2I432.1.35 Control System Subtype	1	Fixed	0: N/A 1: Integrated HOT 2: Standalone HOT

Example of Loco ID Record:

```
2011/06/05|12:00:00.000|20|LID|8|1|RR|6006|RR|120|4|L|132|6|R|400|90|6000|AC|6|42.00|20|1|1|1|1|0|0|0|0|2|1|1|1|1|1|0
```

Note: Shaded: record header, Non-shaded: data record

3.22 EMP Message (EMP) – Version 2

L2I434

Log and record the data record when a BOS or MDM message is transmitted or received by the I-ETMS On-board Application.

L2I433

MDM messages include MDM Server to MDM Client and MDM Client to I-ETMS On-board Application.

L2I877006

Log the data record when an Energy Management message is transmitted or received by the I-ETMS On-board Application according to the requirement in the I-ETMS On-Board Segment Requirement Specification. Logging of this information is subject to common and railroad specific configuration settings.

Record Content:

L2I434.1 Field	Max Size (Bytes)	Type Text Fixed/Variable	Description
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		Size	
L2I434.1.2 Message ID	5	Variable	nnnnn
L2I434.1.3 Packet Number	2	Variable	Packet Number
L2I434.1.4 Total Packet	2	Variable	Total Packets comprising EMP Message
L2I434.1.5 Message	16000	Variable	Hexadecimal format of EMP header, message body and DIV

Example of EMP Message Record:

2011/06/05|12:00:00.000|21|EMP|2|02005|1|4|0fca45...

Note: Shaded: record header, Non-shaded: data record

3.23 EMP Message Failure (EMF) – Version 3

L2I435

Log and record the data record when an Office or WIU message is rejected or failed to receive a response.

L2I877007

Log the data record when an Energy Management message is rejected. Logging of this information is subject to common and railroad specific configuration settings.

Record Content:

L2I435.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I435.1.1 Message ID	5	Variable	nnnnn
L2I435.1.2 Message Failure	1	Fixed	0: HMAC Failed 1: CRC Failed 2: Invalid 3: Stale 4: Duplicate 5: Incorrect State 6: Rejected 7: Ack not Received

Example of EMP Message Failure Record:

2011/06/05|12:00:00.000|22|EMF|3|01005|6

Note: Shaded: record header, Non-shaded: data record

3.24 Track Data Version (TDV) – Version 3

L2I469

Log and record the data record of each track subdiv installed when a new track version is installed.

Record Content:

L2I469.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I469.1.5 Number of Subdiv Files Installed	2	Variable	1 – 99, record should contain at least one sub/dist. Repeat the following fields for each number of Sub/Dist installed.
L2I469.1.10 SCAC	4	Variable	SCAC of the Sub/Dist
L2I469.1.7 Sub/Dist ID	5	Variable	ID of Sub/Dist
L2I469.1.8 Track Data Version	5	Variable	Version of Sub/Dist
L2I469.1.9 Track Data CRC	8	Fixed	CRC-32.
L2I469.1.11 Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown

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Example of Track Data Version Record:

2011/06/05|12:00:00.000|23|TDV|3|3|RR|201|1|d7500316|RR|210|2|f6520de3|RR|250|1|34687ac3|US

Note: Shaded: record header, Non-shaded: data record

3.25 Exclusion Zone (EXC) - Version 2

L2I437

Log and record the data record when an exclusion zone is created or discarded.

Record Content:

L2I437.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I437.1.1 Exclusion Zone Status	1	Fixed	C: Created D: Discarded
L2I437.1.2 Start Location Operating Railroad SCAC	4	Variable	SCAC of the operating railroad of the subdivision of the exclusion zone's starting location
L2I437.1.3 Start location Subdivision/District ID	5	Variable	Track subdivision/district ID of the exclusion zone's starting location
L2I437.1.4 Start Location Block ID	10	Variable	Track segment block ID of the exclusion zone's starting location
L2I437.1.5 Start Location Offset	6	Variable	Foot offset into block of the exclusion zone's starting location
L2I437.1.10 Start Location Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown
L2I437.1.6 End Location Operating Railroad SCAC	4	Variable	SCAC of the operating railroad of the subdivision of the exclusion zone's ending location
L2I437.1.7 End Location Subdivision/District ID	5	Variable	Track subdivision/district ID of the exclusion zone's ending location
L2I437.1.8 End Location Block ID	10	Variable	Track segment block ID of the exclusion zone's ending location
L2I437.1.9 End Location Offset	6	Variable	Foot offset into block of the exclusion zone's ending location
L2I437.1.11 End Location Country of Operation	2	Fixed	US: US CA: Canada UK: Unknown

Example of Exclusion Zone Record:

2011/06/05|12:00:00.000|27|EXC|2|C|RR|321|1103|10050|US|RR|321|1104|65|US

Note: Shaded: record header, Non-shaded: data record.

3.26 Energy Management Control Indicator (ECI) – Version 2

L2I543876

Log and record the data record when the Energy Management Control indicator changes state.

Record Content:

L2I543876.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I543876.1.1 Indicator Status	1	Fixed	D: Displayed; log when indicator first appears or reappears R: Removed; log when indicator is no longer displayed
L2I543876.1.2 Source Code	3	Fixed	PTC: Positive Train Control

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			EMS: Energy Management System
L2I543876.1.3 Indicator Text	15	Variable	Text to display in the control indicator

Example of Energy Management Control Indicator Record:

2011/06/05|12:00:00.000|28|ECI|2|D|EMS|EMS

Note: Shaded: record header, Non-shaded: data record.

3.27 Energy Management Overlay Display (EOD) – Version 2

L2I543889

Log and record the data record when the Energy Management Overlay changes state.

Record Content:

L2I543889.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I543889.1.1 Indicator Status	1	Fixed	D: Displayed; log when indicator first appears or reappears R: Removed; log when indicator is no longer displayed
L2I543889.1.2 Source Code	3	Fixed	PTC: Positive Train Control EMS: Energy Management System
L2I543889.1.3 Overlay Line 1	16	Variable	First line of a top 2-line or first line of a 5-line ITF/BCP Overlay text as displayed
L2I543889.1.4 Overlay Line 2	16	Variable	Second line of a top 2-line or second line of a 5-line ITF/BCP Overlay text as displayed
L2I543889.1.5 Overlay Line 3	16	Variable	Third line of a 5-line ITF/BCP Overlay text as displayed, or NULL if no 5-line text
L2I543889.1.6 Overlay Line 4	16	Variable	First line of a bottom 2-line or fourth line of a 5-line ITF/BCP Overlay text as displayed
L2I543889.1.7 Overlay Line 5	16	Variable	Second line of a bottom 2-line or fifth line of a 5-line ITF/BCP Overlay text as displayed

Example of Energy Management Overlay Display Record:

2011/06/05|12:00:00.000|29|EOD|2|D|EMS|Line 1|Line 2|Line 3|Line 4|Line 5

Note: Shaded: record header, Non-shaded: data record.

3.28 Energy Management Speed Prediction (ESP) – Version 2

L2I543900

Log and record the data record when the Energy Management Speed Prediction changes state.

Record Content:

L2I543900.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I543900.1.1 Indicator Status	1	Fixed	D: Displayed; log when indicator first appears or reappears R: Removed; log when indicator is no longer displayed
L2I543900.1.2 Source Code	3	Fixed	PTC: Positive Train Control EMS: Energy Management System
L2I543900.1.3 Number of Speed Predictions	1	Fixed	0-12
L2I543900.1.4 Speed / Text	10	Variable	Speed or text displayed on track map
L2I543900.1.5	4	Variable	Railroad SCAC of the railroad

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Railroad SCAC			subdivision/district where text is to be displayed
L2I543900.1.6 Subdivision/District ID	5	Variable	Track Subdivision/District ID where text is to be displayed
L2I543900.1.7 Track Segment ID	10	Variable	Track Segment Block ID where text is to be displayed
L2I543900.1.8 Offset into track segment	6	Variable	Feet into track segment
L2I543900.1.9 Blink	1	Fixed	0 = Don't blink on display 1 = Blink on display

Example of Energy Management Speed Prediction Record:

2011/06/05|12:00:00.000|30|ESP|2|D|EMS|3|Text 1|RR|321|1104|65|1|Text 2|RR|321|1105|165|0|Text 3|RR|321|1106|665|0

Note: Shaded: record header, Non-shaded: data record.

3.29 Energy Management Information (EMI) - Version 2

L2I612852

Log and record the data record when either:

- a data field changes status or
- any Energy Management Status (0600) EM message is received during EM interface negotiation until a match between the Interface Version field in the PTC Status Data (0500) and Energy Management Status (0600) messages is achieved, or no prior interface version existed.

Record Content:

L2I612852.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I612852.1.1 On-Board - EM ICD Interface Version requested by On-Board PTC application	1	Variable	Interface version sent in the 0500 message to EM
L2I612852.1.2 On-Board - EM ICD Interface Version supported by Energy Management	1	Variable	Interface version received in the 0600 message from EM
L2I612852.1.3 Requested Interface Version matched a version On-board	1	Fixed	0: No 1: Yes
L2I612852.1.4 Energy Management Advisory Functions Ready	1	Fixed	0: No 1: Yes
L2I612852.1.5 Energy Management Cruise Functions Ready	1	Fixed	0: No 1: Yes
L2I612852.1.6 Suppress EM Access Screen Soft-Key Display	1	Fixed	0: No 1: Yes
L2I612852.1.7 Energy Management Advisory functions active	1	Fixed	0: No 1: Yes
L2I612852.1.8 Energy Management Cruise functions active	1	Fixed	0: No 1: Yes
L2I612852.1.9 Energy Management Cruise function exit initiated	1	Fixed	0: No 1: Yes

Example of Energy Management Information Record:

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2020/06/05|12:00:00.000|31|EMI|2|1|5|5|1|1|1|1|1|1

Note: Shaded: record header, Non-shaded: data record.

3.30 Serial Cab Signal Interface (CSI) - Version 1

L2I651012

Log and record the data record when Serial Cab Signal System Interface status changes and the Serial Cab Signal System Interface is connected.

Record Content:

L2I651012.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I651012.1.1 PTC Status	1	Fixed	0: Not Active 1: Active
L2I651012.1.2 Cab Signal Status	1	Fixed	0: Unknown 1: Higher Speed Operation Disabled 2: Higher Speed Operation Enabled
L2I651012.1.3 Connection State	1	Fixed	0: Disconnected 1: Connected

Example of Serial Cab Signal Interface Record:

2011/06/05|12:00:00.000|32|CSI|1|0|0|0

Note: Shaded: record header, Non-shaded: data record.

3.31 EOT GPS (EGP) - Version 1

L2I769862

Log and record the data record when received from the EOT via the HOTCC.

Record Content:

L2I769862.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I769862.1.1 GPS Messages	1	Fixed	1: NMEA 2: UBX-NAV-POSECEF
L2I769862.1.2 Position Fix	1	Variable	0: Invalid 1: Valid SPS/Standalone 2: Valid DGPS/SBAS 3: Valid PPS 4: RTK Note: Invalid if GPS Message is UBX-NAV-POSECEF.
L2I769862.1.3 Latitude	10	Variable	dd:mm.mmmm Note: field empty if GPS Message is UBX-NAV-POSECEF.
L2I769862.1.4 N/S Indicator	1	Fixed	I: Invalid N: North S: South Note: Invalid if GPS Message is UBX-NAV-POSECEF.
L2I769862.1.5 Longitude	11	Variable	ddd:mm.mmmm Note: field empty if GPS Message is UBX-NAV-POSECEF.
L2I769862.1.6 E/W Indicator	1	Fixed	I: Invalid E: East W: West Note: Invalid if GPS Message is UBX-NAV-

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			POSECEF.
L2I769862.1.7 ECEF_X coordinate	10	Variable	Centimeters The X axis is the plane of the equator, passing through extending from 90°W longitude (negative) to the prime meridian (positive) Note: field empty if GPS Message is NMEA.
L2I769862.1.8 ECEF_Y coordinate	10	Variable	Centimeters The Y axes is also in the plane of the equator, passing through extending from 90°W longitude (negative) to 90°E longitude (positive) Note: field empty if GPS Message is NMEA.
L2I769862.1.9 ECEF_Z coordinate	10	Variable	Centimeters The Z axis is the line between the North and South Poles with positive values increasing northward. Note: field empty if GPS Message is NMEA.
L2I769862.1.10 Position Accuracy Estimate	4	Variable	0-5000 Centimeters Note: field empty if GPS Message is NMEA.
L2I769862.1.11 Heading GPS	5	Variable	1-360: Valid 65535: Invalid
L2I769862.1.12 Number of Satellites	2	Fixed	00 - 99, total number of satellites across all constellations
L2I769862.1.13 HDOP	4	Variable	nn.n

Example of EOT GPS Record:

2011/06/05|12:00:00.000|33|EGP|1|1|2|33:24.7020|N|80:50.5514|W|30|07|1.4

Note: Shaded: record header, Non-shaded: data record.

3.32 EOT Status (EST) - Version 1

L2I769873

Log and record the data record when received from the EOT via the HOTCC.

Record Content:

L2I769873.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I769873.1.1 EOT ID	5	Variable	EOT ID of the EOT for the train 0 – EOT Unlinked 1 to 99,999 – EOT ID
L2I769873.1.2 Propagation Threshold Time	14	Fixed	UTC Time when Air Propagation Threshold crossed YYYYMMDDHHMMSS
L2I769873.1.3 UTC Source	10	Fixed	0 – GPS 1 – Cell tower 2 – RTC
L2I769873.1.4 Brake Pipe Pressure	3	Variable	PSI 1 to 127 – Valid range 255 – Unknown
L2I769873.1.5 EOT Mode	3	Variable	1 – Air Run Mode 2 – Battery Run Mode 3 – Charger Mode 4 – Sleep and Tracking Mode 5 – Ship Mode 255 – Unknown
L2I769873.1.6 EOT Enable State	3	Variable	1 – Enabled 2 – Pending Disabled

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			3 – Disabled 255 – Unknown
L2I769873.1.7 Valve Status	3	Variable	Valve Status 0 – valve fail 1 – valve good 255 – Unknown
L2I769873.1.8 Motion Status	3	Variable	Status of Motion 0 – stopped/vertical 1 – moving vertical 3 – horizontal 254 – failed motion detector 255 – unknown
L2I769873.1.9 Marker Status	3	Variable	Status of Marker Light 0 – off 1 – on 3 – failed marker light 255 – Unknown
L2I769873.1.10 Generator Voltage	4	Variable	Output of the Generator Scaled in 0.1 VDC per bit 25.5 – Unknown
L2I769873.1.11 Battery Voltage	6	Variable	Voltage of Battery Scaled in Millivolts 0-16.000 = Valid 65.535 – Unknown
L2I769873.1.12 Battery Charge	3	Numeric	Current Battery Charge 0 to 100 – % 255 – Unknown
L2I769873.1.13 RSSI	5	Variable	Received Signal Strength of Last UHF message. Cleared 2 minutes after last message. Raw signed dBm value 32767 – Unknown
L2I769873.1.14 Seconds Since Last UHF Message	5	Variable	Seconds since Last UHF Message received 0 to 900 Seconds 65535 – Unknown

Example of EOT Status Record:

2011/06/05|12:00:00.000|34|EST|1|12345|20230523090000|2|90|1|1|1|0|1|12.5|13.000|95|95|15

Note: Shaded: record header, Non-shaded: data record.

3.33 HOTCC Data Status (HDS) - Version 1

L2I769884

Log and record the data record when received from the EOT via the HOTCC.

Record Content:

L2I769884.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I769884.1.1 EOT Pair Status	1	Fixed	0 – Not Paired 1 – Pairing 2 – Paired (PTC EOT) 3 – Paired Non PTC 4 – Unpairing 5 – Pair Failed 6 – Paired to Other Locomotive
L2I769884.1.2 EOT Comm State	1	Fixed	0 – Not Communicating 1 – Communicating
L2I769884.1.3 EOT Camera	1	Fixed	0 – EOT does not have camera

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Equipped			1 – EOT has camera
L2I769884.1.4 EOT Camera State	1	Fixed	0 – Off 1 – Slow Rate 2 – Fast Rate
L2I769884.1.5 EOT ID	5	Variable	EOT ID of the EOT for the train 0 – EOT Unlinked 1 to 99,999 – EOT ID
L2I769884.1.6 Other Locomotive ID	10	Fixed	Locomotive ID of the Other Locomotive All spaces – Invalid
L2I769884.1.7 HOT Link Status	1	Fixed	0 – Not Linked 1 – HOT Linked
L2I769884.1.8 HOT Comm State	1	Fixed	0 – Not Communicating 1 – Communicating

Example of HOTCC Data Status Record:

2011/06/05|12:00:00.000|35|HDS|1|0|0|0|12345| |1|1

Note: Shaded: record header, Non-shaded: data record.

3.34 Audible Alert (AUD) - Version 1

L2I961292

The on-board segment shall log the AUD data record as defined in the following table when an audible alert occurs.

L2I961302

The on-board segment shall record the AUD data record as defined in the following table when an audible alert occurs.

Record Content:

L2I961292.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I961292.1.1 Number of Beeps	1	Fixed	Number of beeps sounded: 1, 2, or 3
L2I961292.1.2 Beep Intensity	1	Fixed	0 = Soft 1 = Loud
L2I961292.1.3 Reason for Alert	2	Fixed	01: System Failure 02: On-board State Change 03: System Prompt 04: Non-Enforceable Warning 05: Predictive/Reactive Warning 06: Braking in Progress 07: Audible Alert Test 08: Energy Management 09: Full Page Banner 10: View All Prompts Key

Example of Audible Alert Record:

2011/06/05|12:00:00.000|39|AUD|1|3|1|06

Note: Shaded: record header, Non-shaded: data record

3.35 Menu Selection (MNU) - Version 1

L2I961305

The on-board segment shall log the MNU data record as defined in the following table when a key is pressed to select a line of text in a menu.

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L2I961306

The on-board segment shall record the MNU data record as defined in the following table when a key is pressed to select a line of text in a menu.

Record Content:

L2I961305.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I961305.1.1 Key Text	30	Variable	Text of pressed key
L2I961305.1.2 Selected Text	80	Variable	Text of selected item from menu list

Example of Menu Selection Record:

2011/06/05|12:00:00.000|40|MNU|1|Select|RR

Note: Shaded: record header, Non-shaded: data record.

3.36 Train Handling Data (THD) - Version 1

L2I1076820

Log the data record every second when TDL-CFG1 is enabled.

Note: This train handling data reflects the raw values received from the control system and may not match the conditioned values used for PTC functions.

Record Content:

L2I1076820.1 Field	Max Size (Bytes)	Type Text Fixed/Variable Size	Description
L2I1076820.1.1 DIO Brake Pipe Pressure 1 (raw from control system)	3	Variable	psi 255 - Invalid
L2I1076820.1.2 DIO Brake Pipe Pressure 2 (raw from control system)	3	Variable	psi 255 - Invalid
L2I1076820.1.3 ILC Brake Pipe Pressure (raw from control system)	3	Variable	psi 255 - Invalid
L2I1076820.1.4 Brake Cylinder Pressure (raw from control system)	3	Variable	psi 255 - Invalid
L2I1076820.1.5 Dynamic Brake Excitation (raw from control system)	5	Variable	VDC 65535 - Invalid
L2I1076820.1.6 Throttle Notch (raw from control system)	3	Variable	1 to 10 0 – Idle 255 - Invalid

Example of Train Handling Data Record:

2026/06/05|12:00:00.000|41|THD|1|90|90|90|0|127|0

Note: Shaded: record header, Non-shaded: data record.

3.37 Phase Break (PBK) - Version 1

L2I594570

Log and record the data record when the phase break state changes

Record Content:

L2I594570.1 Field	Max Size	Type Text Fixed/Variable	Description
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	(Bytes)	Size	
L2I594570.1.1 Phase Break	1	Variable	0 = Phase Break Released 1 = Phase Break Activated
L2I594570.1.2 Demand Setting	6	Variable	% Demand Setting
L2I594570.1.3 Power (PCS)	1	Variable	0 = Power available 1 = Power removed
L2I594570.1.4 Start Location Operating Railroad SCAC	4	Variable	SCAC of Operating Railroad of the subdiv of the phase break's starting location
L2I594570.1.5 Start Location Sub/Dist ID	5	Variable	Track Subdivision / District ID of the phase break's starting location
L2I594570.1.6 Start Location Block ID	10	Variable	Track Segment Block ID of the phase break's starting location
L2I594570.1.7 Start Location Offset	6	Variable	Foot offset into Block of the phase break's starting location
L2I594570.1.8 End Location Operating Railroad SCAC	4	Variable	SCAC of Operating Railroad of the subdiv of the phase break's ending location
L2I594570.1.9 End Location Sub/Dist ID	5	Variable	Track Subdivision / District ID of the phase break's ending location
L2I594570.1.10 End Location Block ID	10	Variable	Track Segment Block ID of the phase break's ending location
L2I594570.1.11 End Location Offset	6	Variable	Foot offset into Block of the phase break's ending location

Example of Phase Break Record:

2011/06/05|12:00:00.000|37|PBK|1|1|30|0|RR|321|1103|10050|RR|321|1104|14520

Note: Shaded: record header, Non-shaded: data record.

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